

Updates on data processing pipeline

23 December 2025

Key Changes

- Patch extraction was rebuilt: positives come from the malignant mask or its bounding box with a minimum lesion coverage, negatives are split into near/far groups with coverage caps, intensity filters, and balanced 1:1 targets for malignant vs. healthy patches.
- Patient-level train/val/test splitting is now supported (seeded) to avoid leakage, with per-split patch folders and patient IDs carried through loading and feature building.
- Config, selection, and models with support for the new sampling features (coverage thresholds, near/far radii, nonzero/intensity filters) plus group subsampling and GroupShuffleSplit based validation; mean/std are only stored when standardizing and AUC handling is more robust.
- Pipeline now loads arbitrary splits, to keep patient labels through response computation, evaluates trained models on additional splits, and logs per split metrics alongside ROC/PR and confusion plots.
- Drop patches containing dark pixels more than 35%.



Figure 1: Inspection of data.

I was expecting more changes in decision boundary graph but I got a similar pattern. I ran multiple experiments with different configuration of kernel selection (ABESS/MMR, Number of Kernels, K best kernels) but unfortunately I could not get a better distinction.

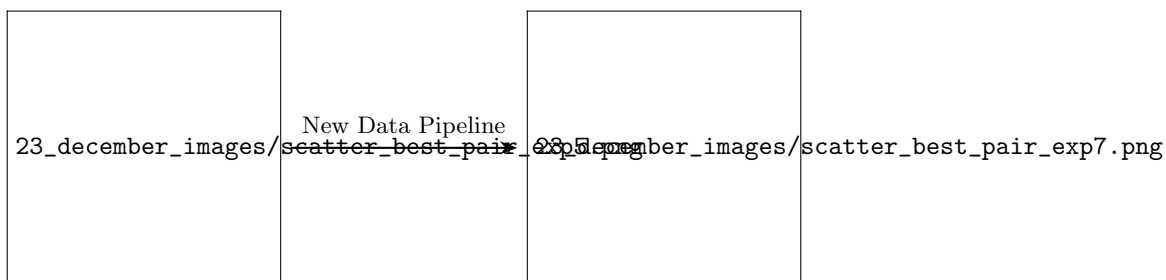


Figure 2: Overview of the new data pipeline from original samples to processed outputs.

More changes and a new experiment

- In the previous experiments, I was using the non malignant parts of the image (healthy parts of malignant image) as healthy, in the following I included patches from truly healthy data as well.
- Feature side: per-patch responses are z-scored before convolution and default aggregation switched to `abs_max`. (I think not a good idea)
- Standardize patches using their mean and std.

I managed to make a less reliable experiment using these 3 changes:



Figure 3:

Current plans:

- Rebalance negatives: reduce/disable malignant-background negatives and keep most negatives from healthy slides; also increase far neg radius.
- Loosen positive strictness slightly: drop `min_pos_coverage` back toward 0.6–0.7; keep `bbox` positives off (bbox around the lesion tissue).
- Try `response_fn="abs_max"` vs `mean_abs` and turn off patch z-scoring.
- Model: standardize features, use logistic (not MLP) for the 2D/3D plots to avoid overfitting on small subsets; rerun subset search with a fixed `train_subset_frac` (e.g., 0.5) for stability.

- And a lot more experiments...

[link to github](#)